

LECTURE-7

Wednesday
9/9/26.

□ Quick Sort: (Divide & Conquer).

↳ Inplace

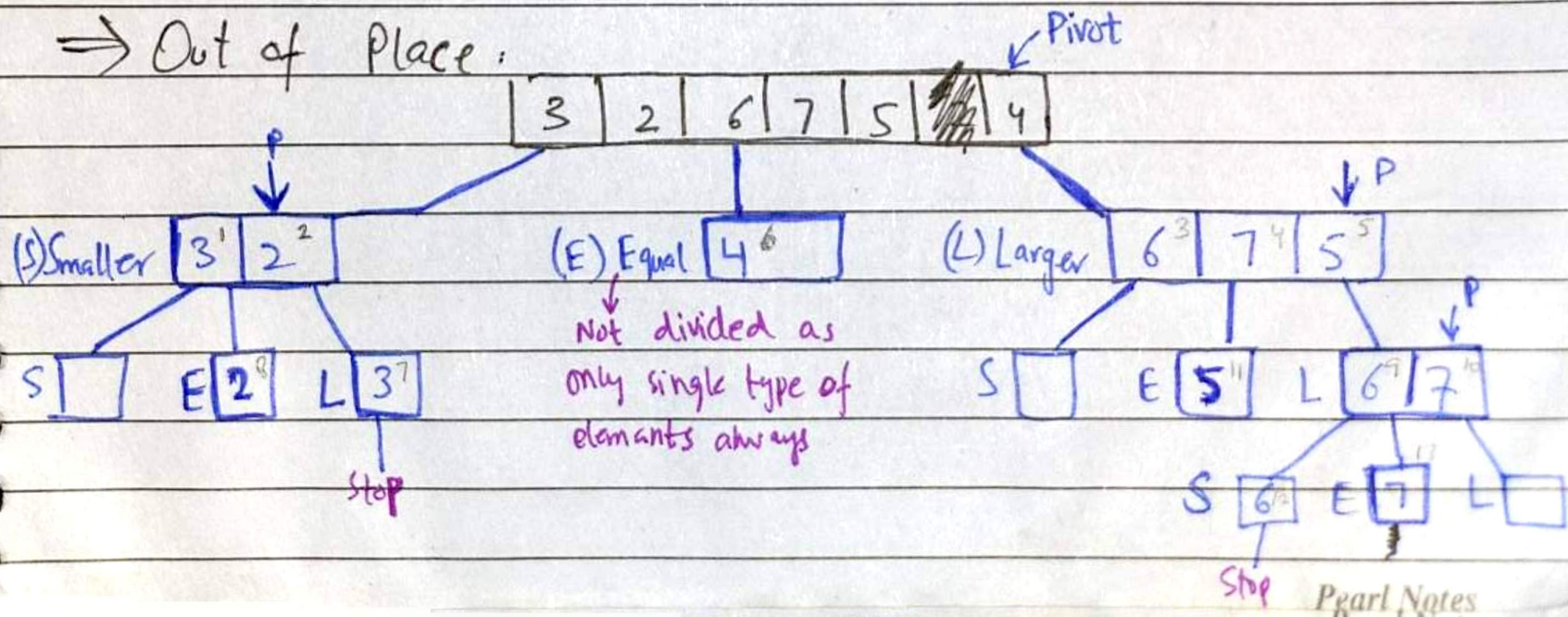
↳ Out of place

Best: $n \log n$

Worst: n^2 (when data already sorted).

Pivot element: The last data element which won't be considered as a part of data & is crucial for sorting.

⇒ Out of Place:



Explanation: We'll have a pivot at last & then we'll keep placing elements of original array in 3 arrays. Smaller than pivot goes in 'S'; larger than pivot goes in 'L' & equal goes in 'E'. After one iteration; do the same for each of the 'S' and 'L' array i.e. choose pivot & further create S, E, L arrays. When only one element in ~~L~~ L/S & other one is empty; we'll stop. Then after we've each branch sorted in order; then we'll just combine the arrays in S → E → L order which will sort the original array.

$\boxed{2} \boxed{3} + \boxed{4} + \boxed{5} \boxed{6} \boxed{7} = 2, 3, 4, 5, 6, 7$
from merge sort

Best: $T(n) = 2T(n/2) + \mathcal{O}(n) \longrightarrow T(n) = \mathcal{O}(n \log n)$

Worst: $T(n) = T(n-1) + T(0) + \mathcal{O}(n) \longrightarrow T(n) = \mathcal{O}(n^2)$

⇒ InPlace: we'll have two pointers and pivot will be last element.

i points end of 'small' array (& start of 'large')

x points the current element & act as a global moving iterator.

- The goal is to swap the data with i+1 if the iterator 'i' gets on an element that is smaller/equal to pivot's data.

Else if the data x points is larger than pivot; then we'll just move x. (Only x can move OR x & i both will move).

QuickSort (A [1:n], l, h) {

if (l ≥ h) {return}

else {

 pivot = partition (A, l, h)

 QuickSort (A, l, pivot-1) // Smaller

 QuickSort (A, pivot+1, h) // Larger

}

}

Indices: 1 2 3 4 5 6 ^{pivot} 7 Date _____

3	2	6	7	5	1	4
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- $i=0, x=0 \rightarrow i=1, x=1 [3 \leq \text{pivot}(4)] \rightarrow i=2, x=2 [2 \leq \text{pivot}]$
 $i=2(\text{end of small}), x=4 [7 > \text{pivot}] \leftarrow i=2, x=3 [6 > \text{pivot}] \leftarrow$
 $\rightarrow i=2, x=5 [5 > \text{pivot}] \rightarrow i=3, x=6$ and now we'll swap $A[i+1]$ (i.e.: 6) with $A[x]$ (i.e.: 1) so new array: ~~3, 2, 1, 7, 5, 6, 4~~
 $3, 2, 1, 7, 5, 6, 4 \rightarrow i=4, x=7 [4 \leq \text{pivot}]$ so now swap '4' with '7' and we'll have new array: 3, 2, 1, 4, 5, 6, 7

End Product:

3	2	1	4	5	6	7
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smaller pivot larger

*Now we can refer the end of smaller array as "p-1" & start of larger array as p+1